

Summation via Contour Integration

Evaluating $\sum_{n=-\infty}^{\infty} \frac{1}{n^2 + a^2}$

Complex Analysis

Goal

We compute the sum

$$S = \sum_{n=-\infty}^{\infty} \frac{1}{n^2 + a^2}, \quad a > 0.$$

This will be done using contour integration.

The final result will be

$$S = \frac{\pi}{a} \coth(\pi a).$$

Key Idea

We use the function

$$f(z) = \frac{\pi \cot(\pi z)}{z^2 + a^2}.$$

Why this choice?

- ▶ $\cot(\pi z)$ has simple poles at $z = n \in \mathbb{Z}$
- ▶ Residue at $z = n$ is $1/\pi$
- ▶ Thus $\pi \cot(\pi z)$ has residue 1 at each integer

This will allow us to recover the sum.

Poles of the Integrand

The function

$$f(z) = \frac{\pi \cot(\pi z)}{z^2 + a^2}$$

has poles at:

- ▶ $z = n \in \mathbb{Z}$ (from $\cot(\pi z)$)
- ▶ $z = ia, \quad z = -ia$ (from denominator)

Residues at the Integers

Near $z = n$,

$$\pi \cot(\pi z) \sim \frac{1}{z - n}.$$

Thus

$$\operatorname{Res}(f, n) = \frac{1}{n^2 + a^2}.$$

Therefore

$$\sum_{n=-\infty}^{\infty} \operatorname{Res}(f, n) = \sum_{n=-\infty}^{\infty} \frac{1}{n^2 + a^2}.$$

Contour Choice

We integrate $f(z)$ over a large rectangle:

- ▶ vertices at $\pm(N + \frac{1}{2}) \pm iR$

This contour encloses all integers n with $|n| \leq N$.

We let

$$N \rightarrow \infty, \quad R \rightarrow \infty.$$

Behavior on the Contour

We must show

$$\int_C f(z) dz \rightarrow 0.$$

Key facts:

- ▶ $\cot(\pi z)$ is bounded on vertical lines
- ▶ denominator grows like $|z|^2$

Thus

$$f(z) = O\left(\frac{1}{|z|^2}\right)$$

and the contour integral vanishes.

Applying Residue Theorem

By the residue theorem:

$$\oint_C f(z) dz = 2\pi i \sum \text{Res}(f, \text{poles}).$$

Since the integral vanishes,

$$0 = 2\pi i \sum \text{Res}.$$

Thus

$$\sum_{\text{all residues}} = 0.$$

Separating the Contributions

We split residues into:

- ▶ integers n
- ▶ poles at $z = \pm ia$

Thus

$$\sum_{n=-\infty}^{\infty} \frac{1}{n^2 + a^2} + \text{Res}(f, ia) + \text{Res}(f, -ia) = 0.$$

Residue at $z = ia$

$$\operatorname{Res}(f, ia) = \lim_{z \rightarrow ia} \frac{\pi \cot(\pi z)}{z + ia}.$$

Thus

$$\operatorname{Res}(f, ia) = \frac{\pi \cot(\pi ia)}{2ia}.$$

Evaluate $\cot(\pi ia)$

Recall

$$\cot(ix) = i \coth(x).$$

Thus

$$\cot(\pi ia) = i \coth(\pi a).$$

Therefore

$$\operatorname{Res}(f, ia) = \frac{\pi i \coth(\pi a)}{2ia} = \frac{\pi}{2a} \coth(\pi a).$$

Residue at $z = -ia$

Similarly,

$$\operatorname{Res}(f, -ia) = \frac{\pi}{2a} \coth(\pi a).$$

Thus total contribution:

$$\operatorname{Res}(ia) + \operatorname{Res}(-ia) = \frac{\pi}{a} \coth(\pi a).$$

Final Result

From

$$\sum_{n=-\infty}^{\infty} \frac{1}{n^2 + a^2} + \frac{\pi}{a} \coth(\pi a) = 0$$

we obtain

$$\sum_{n=-\infty}^{\infty} \frac{1}{n^2 + a^2} = \frac{\pi}{a} \coth(\pi a).$$

Summary

Key ideas:

- ▶ use $\pi \cot(\pi z)$ to generate integer poles
- ▶ build a function whose residues encode the sum
- ▶ use a large contour with integral pieces that decay.
- ▶ compute remaining residues explicitly

This technique is widely used to evaluate infinite sums.